

Designing a Scalable Serverless Event-Driven Architecture on AWS for an E-Commerce Platform

Client Business Overview:

The client has an E-Commerce business. They sell their products all over the world through their website. They have a dedicated “order service” that collects all the orders from across various frontends. The “order service” is hosted on-premises, which validates, authenticates, accepts, processes, and stores orders in the database (MySQL), which consists of one big table. The order service then triggers the other downstream services, like inventory, fulfilment, and accounting.

Issues Facing:

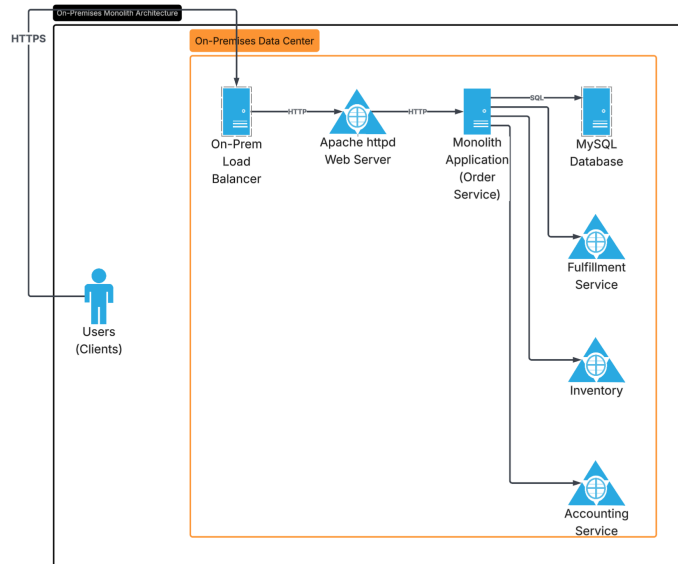
- Business logic is all tied together in one “Order Service”.
- When overloaded, the “Order Service” slows down and sometimes crashes.
- Orders fail when the application is overwhelmed.
- Facing a hard time scaling the system fast enough on premises.
- When the system is down, the “Order Service” does not get the chance to send the information to downstream services for successful orders.
- When an API call fails, it creates an inconsistency in the order data.
- Facing a big challenge when there are traffic spikes due to sales, coupons, and promotions.

Requirements:

- **Host web backend infrastructure on AWS** to accept and process orders.
- Use a **database in AWS** that is highly available, durable, and easier to manage than the current MySQL on-premises.
- Implement **automatic scaling** to handle spiky demand, scaling up during high traffic and scaling down when demand is low.
- **Decouple components** so that downstream calls (e.g., inventory, accounting) are independent and don’t affect order processing.
- Ensure **monitoring and logging** are easy to set up and use a unified logging system.
- **Optimise for cost and performance** to make the solution efficient and affordable.
- Prefer a **serverless or managed** approach to reduce operational overhead.

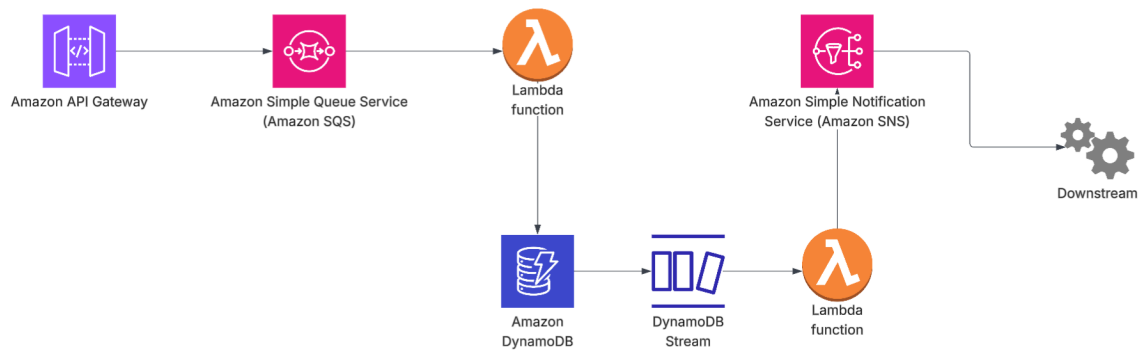
Current Architecture:

Carefully analyzing the business logic and description, we can visualize the current architecture of the existing system as shown in the following:



Proposed Solution:

To address the requirements provided, I designed a serverless, event-driven architecture. The summary goes as follows:



Amazon API Gateway: Will act as a front door, receiving all requests/traffic from the frontend (mobile app, desktop, etc.). It authenticates incoming requests and validates their format before passing them on.

Amazon SQS (Simple Queue Service): Serves as a message queue that holds incoming order requests temporarily. This decouples the API from the processing logic, allowing smooth handling of traffic spikes.

AWS Lambda: Runs the application code to process orders. One Lambda function reads messages from the SQS queue, processes the orders, and stores them in DynamoDB. Another Lambda function processes DynamoDB streams to send order updates downstream.

Amazon DynamoDB: A fast, scalable NoSQL database that stores order data.

DynamoDB Streams: Captures changes in the DynamoDB table (like new orders) and triggers the second Lambda function to handle downstream notifications.

Amazon SNS (Simple Notification Service): Publishes order information to multiple subscribed services (fulfilment, accounting, inventory) using a fan-out pattern.

Amazon CloudWatch logs: Collects monitoring data and logs from all services to help track performance and troubleshoot issues.

AWS Services Used

- IAM
- Lambda
- CloudWatch Logs
- API Gateway
- DynamoDB
- DynamoDB Streams
- SQS
- SNS

Security Considerations

Security best practices were integrated throughout the architecture design:

- IAM least-privilege access policies
- Controlled API access
- Secure service-to-service permissions
- Centralized monitoring and logging
- Managed infrastructure reducing attack surface

Business Outcome Matrix:

- Reduced operational overhead
- Improved scalability
- Minimised failure propagation
- Improved resilience
- Lower infrastructure management burden

Cost Optimization:

- Pay-per-use Lambda
- Serverless scaling

- Reduced infrastructure maintenance
- No server management
- DynamoDB on-demand scaling

Final Thoughts

This project demonstrated how modern AWS serverless architectures can improve scalability, reliability, operational efficiency, and fault tolerance for high-traffic business applications.

Building this solution also helped deepen my understanding of cloud-native design patterns, distributed systems, and AWS architecture best practices.

#AWS #CloudComputing #Serverless #Lambda #DevOps #Terraform #CloudArchitecture
#DynamoDB

Implementation Process

Following is the task-by-task process of creating a proof of concept with screenshots:

- Create IAM policies and roles to follow best practices of working in the AWS Cloud.
- Create a DynamoDB table to store data.
- Create an Amazon SQS queue to receive, store, and send messages between software components.
- Create Lambda functions and set up triggers to invoke actions in different AWS services.
- Enable DynamoDB Streams to capture modifications in the database table.
- Configure Amazon SNS to receive email or text notifications.
- Create a REST API to insert data into a database.

Task 1. Setup: Security & Access Management Design

Task 1.1: Creating custom IAM policies

Lambda-Write-DynamoDB

The screenshot displays the AWS IAM console interface for creating a new policy. The breadcrumb navigation shows 'IAM > Policies > Create policy'. The left sidebar indicates the current step is 'Specify permissions' (Step 1) and 'Review and create' (Step 2).

Specify permissions [Info](#)

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Policy editor

Visual | **JSON** | Actions | [Icon]

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "VisualEditor0",
6       "Effect": "Allow",
7       "Action": [
8         "dynamodb:PutItem",
9         "dynamodb:DescribeTable"
10      ],
11       "Resource": "*"
12     }
13   ]
14 }
```

Edit statement

Select a statement

Select an existing statement in the policy or add a new statement.

[+ Add new statement](#)

Review and create [Info](#)

Review the permissions, specify details, and tags.

Policy details

Policy name
Enter a meaningful name to identify this policy.

Maximum 128 characters. Use alphanumeric and "+,=, @, _" characters.

Description - optional
Add a short explanation for this policy.

Maximum 1,000 characters. Use alphanumeric and "+,=, @, _" characters.

Permissions defined in this policy [Info](#) [Edit](#)

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it

Allow (1 of 469 services) [Show remaining 468 services](#)

Service	Access level	Resource	Request condition
DynamoDB	Limited: Read, Write	All resources	None

Add tags - optional [Info](#)

CloudShell Feedback Console Mobile App

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Lambda-SNS-Publish

Step 1

Specify permissions

Step 2

Review and create

Specify permissions

Info

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Visual

JSON

Actions

Policy editor

1 {

2 "Version": "2012-10-17",

3 "Statement": [

4 {

5 "Sid": "VisualEditor0",

6 "Effect": "Allow",

7 "Action": [

8 "sns:Publish",

9 "sns:GetTopicAttributes",

10 "sns:ListTopics"

11],

12 "Resource": "*"

13 }]

14 }

15 }

Edit statement

Select a statement

Select an existing statement in the policy or add a new statement.

+ Add new statement

Lambda-DynamoDBStreams-Read

Step 1

Specify permissions

Step 2

Review and create

Specify permissions

Info

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

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JSON

Actions

Policy editor

1 {

2 "Version": "2012-10-17",

3 "Statement": [

4 {

5 "Sid": "VisualEditor0",

6 "Effect": "Allow",

7 "Action": [

8 "dynamodb:GetShardIterator",

9 "dynamodb:DescribeStream",

10 "dynamodb:ListStreams",

11 "dynamodb:GetRecords"

12],

13 "Resource": "*"

14 }]

15 }

16 }

Edit statement

Select a statement

Select an existing statement in the policy or add a new statement.

+ Add new statement

Step 1

Specify permissions

Step 2

Review and create

Review and create

Review the permissions, specify details, and tags.

Policy details

Policy name

Enter a meaningful name to identify this policy.

Lambda-DynamoDBStreams-Read

Maximum 128 characters. Use alphanumeric and '+', '@', '-' characters.

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Search

Allow (1 of 469 services)

Show remaining 468 services

Service	Access level	Resource	Request condition
DynamoDB	Limited: Read	All resources	None

Add tags - optional

CloudShell

Feedback

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Lambda-Read-SQS

Step 1

Specify permissions

Step 2

Review and create

Specify permissions

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Policy editor

Visual

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Task 1.2: Creating IAM roles and attaching policies to the roles

IAM

>

Roles

>

Create role

Step 1

●

Select trusted entity

Step 2

●

Add permissions

Step 3

○

Name, review, and create

Select trusted entity

Info

Trusted entity type

☒ AWS service

Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ AWS account

Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ Web identity

Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.

☐ SAML 2.0 federation

Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ Custom trust policy

Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

Lambda

Choose a use case for the specified service.

Use case

CloudShell

Feedback

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IAM

>

Roles

>

Create role

Step 1

●

Select trusted entity

Step 2

●

Add permissions

Step 3

○

Name, review, and create

Add permissions

Info

Permissions policies (2/1152)

Info

Choose one or more policies to attach to your new role.

Filter by Type

Search

lambda-

×

All types

▼

4 matches

< 1 >

⚙

	Policy name ⓘ	Type	Description
☐	Lambda-DynamoDBStreams-Read	Customer managed	-
☒	Lambda-Read-SQS	Customer managed	-
☐	Lambda-SNS-Publish	Customer managed	-
☒	Lambda-Write-DynamoDB	Customer managed	-

► Set permissions boundary - optional

Cancel

Previous

Next

IAM > Roles > Create role

1

Step 1
Select trusted entity

Step 2
Add permissions

Step 3
Name, review, and create

Name, review, and create

Role details

Role name
Enter a meaningful name to identify this role.

Maximum 64 characters. Use alphanumeric and '+*,@,_,-' characters.

Description
Add a short explanation for this role.

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+*, @-/{}#\$\$%^&()-"

Step 1: Select trusted entities

Edit

Trust policy

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Effect": "Allow",  
6       "Action": [  
7         "sts:AssumeRole"
```

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IAM > Roles > Create role

1

Step 1
Select trusted entity

Step 2
Add permissions

Step 3
Name, review, and create

Role Lambda-SQS-DynamoDB created.

View role

Add permissions info

Permissions policies (2/1152) info

Choose one or more policies to attach to your new role.

Filter by Type

All types

4 matches

< 1 >

☐	Policy name info	Type	Description
☒	Lambda-DynamoDBStreams-Read	Customer managed	-
☐	Lambda-Read-SQS	Customer managed	-
☒	Lambda-SNS-Publish	Customer managed	-
☐	Lambda-Write-DynamoDB	Customer managed	-

► Set permissions boundary - optional

Cancel

Previous

Next

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IAM > Roles > Create role

Step 1

Select trusted entity

Step 2

Add permissions

Step 3

Name, review, and create

Name, review, and create

Role details

Role name

Enter a meaningful name to identify this role.

Lambda-DynamoDBStreams-SNS

Maximum 64 characters. Use alphanumeric and `*+=, @-./[]{}#%'^^{}~`` characters.

Description

Add a short explanation for this role.

Allows Lambda functions to call AWS services on your behalf.

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: `_+=, @-./[]{}#%'^^{}~``

Step 1: Select trusted entities

Edit

Trust policy

```
1 - {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "sts:AssumeRole"
8       ]
9     }
10  ]
11 }
```

CloudShell

Feedback

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Cookie preferences

IAM > Roles > Create role

Step 1

Select trusted entity

Step 2

Add permissions

Step 3

Name, review, and create

Select trusted entity Info

Trusted entity type

☒ **AWS service**

Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ **AWS account**

Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ **Web identity**

Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.

☐ **SAML 2.0 federation**

Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ **Custom trust policy**

Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

API Gateway

Choose a use case for the specified service.

Use case

☒ **API Gateway**

Allow API Gateway to push logs to CloudWatch Logs

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Feedback

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IAM > Roles > Create role

Step 1

Select trusted entity

Step 2

Add permissions

Step 3

Name, review, and create

Add permissions Info

Permissions policies (1) Info

The type of role that you selected requires the following policy.

Policy name

▲ | Type ▼

  AmazonAPIGatewayPushToCloudWatchLogs

AWS managed

► Set permissions boundary - optional

Cancel

Previous

Next

- Step 1
● Select trusted entity
- Step 2
● Add permissions
- Step 3
● **Name, review, and create**

Name, review, and create

Role details

Role name

Enter a meaningful name to identify this role.

APIGateway-SQS

Maximum 64 characters. Use alphanumeric and '+=, @- _' characters.

Description

Add a short explanation for this role.

Allows API Gateway to push logs to CloudWatch Logs.

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+=, . @-/[()!#\$%^&*();'":

Step 1: Select trusted entities

[Edit](#)

Trust policy

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "",
6       "Effect": "Allow",
7       "Principal": {
8         "Service": "iam.amazonaws.com"
9       }
10    ]
11  }
```

Task 2: Creating a DynamoDB table

DynamoDB > Tables > Create table

Share your feedback on Amazon DynamoDB

Your feedback is an important part of helping us provide a better customer experience. Take this short survey to let us know how we're doing.

Share feedback

Create table

Table details

info

DynamoDB is a schemaless database that requires only a table name and a primary key when you create the table.

Table name

This will be used to identify your table.

orders

Between 3 and 255 characters, containing only letters, numbers, underscores (_), hyphens (-), and periods (.).

Partition key

The partition key is part of the table's primary key. It is a hash value that is used to retrieve items from your table and allocate data across hosts for scalability and availability.

orderId

String

1 to 255 characters and case sensitive.

Sort key - optional

You can use a sort key as the second part of a table's primary key. The sort key allows you to sort or search among all items sharing the same partition key.

Enter the sort key name

String

1 to 255 characters and case sensitive.

Table settings

Default settings

The fastest way to create your table. You can modify most of these settings after your table has been created. To modify these settings now, choose 'Customize settings'.

Customize settings

Use these advanced features to make DynamoDB work better for your needs.

Default table settings

These are the default settings for your new table. You can change some of these settings after creating the table.

CloudShell

Feedback

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Task 3: Creating an SQS queue

Application Integration

Amazon SQS

A message queuing service

Amazon SQS provides queues for high-throughput, system-to-system messaging. You can use queues to decouple heavyweight processes and to buffer and batch work. Amazon SQS stores messages until microservices and serverless applications process them.

Get started

Learn how to use Amazon SQS by creating a queue, sending a message to the queue, and receiving and processing the message.

[Create queue](#)

Pricing

You can get started with Amazon SQS for free. All customers can make 1 million Amazon SQS requests for free each month. Some applications might be able to operate within this Free Tier limit.

[Cost calculator](#)

Documentation

[Developer guide](#)
[API reference](#)
[FAQs](#)

How it works

```
graph LR; Producers[Producers: Messages are sent from applications, microservices, and other AWS services] --> SQS[Amazon SQS: Queues store messages and wait for consumers to poll]; SQS --> Consumers[Consumers: Messages are processed by applications, Lambda functions, EC2 Instances, and other AWS services];
```

[Amazon SQS](#) > [Queues](#) > Create queue

Create queue

Details

Type

Choose the queue type for your application or cloud infrastructure.

Standard info

At-least-once delivery, message ordering isn't preserved

- At-least-once delivery
- Best-effort ordering

FIFO info

First-in-first-out delivery, message ordering is preserved

- First-in-first-out delivery
- Exactly-once processing

You can't change the queue type after you create a queue.

Name

POC-Queue

A queue name is case-sensitive and can have up to 80 characters. You can use alphanumeric characters, hyphens (-), and underscores (_).

[Amazon SQS](#) > [Queues](#) > Create queue

Access policy

Define who can access your queue.

Choose method

Basic

Use simple criteria to define a basic access policy.

Advanced

Use a JSON object to define an advanced access policy.

Define who can send messages to the queue

Only the queue owner

Only the owner of the queue can send messages to the queue.

Only the specified AWS accounts, IAM users and roles

Only the specified AWS account IDs, IAM users and roles can send messages to the queue.

arn:aws:iam::[redacted]:role/APIGateway-SQS

Define who can receive messages from the queue

Only the queue owner

Only the owner of the queue can receive messages from the queue.

Only the specified AWS accounts, IAM users and roles

Only the specified AWS account IDs, IAM users and roles can receive messages from the queue.

arn:aws:iam::[redacted]:role/Lambda-SQS-DynamoDB

JSON (read-only)

```
{
  "Version": "2012-10-17",
  "Id": "__default_policy_ID",
  "Statement": [
    {
      "Sid": "__owner_statement",
      "Effect": "Allow",
      "Principal": {
        "AWS": "[redacted]"
      },
      "Action": [
        "SQS:*"
      ],
      "Resource": "arn:aws:sqs:us-east-2:[redacted]:POC-Queue"
    },
    {
      "Sid": "__sender_statement",
      "Effect": "Allow",

```

Redrive allow policy - Optional

Identify which source queues can use this queue as the dead-letter queue.

Select which source queues can use this queue as the dead-letter queue.

Disabled

Enabled

Task 4: Order Processing Service Implementation

Task 4.1: Creating a Lambda function for the Lambda-SQS-DynamoDB role

</

Task 4.2: Setting up Amazon SQS as a trigger to invoke the function

Function overview

Diagram | Template

+ Add trigger

POC-Lambda-1

Layers (0)

+ Add destination

Code

Test

Monitor

Configuration

Aliases

Versions

Code source

lambda_function.py

```
1 import json
2
3 def lambda_handler(event, context):
4     # TODO implement
5     return {
6         'statusCode': 200,
7         'body': json.dumps('Hello from Lambda!')
8     }
9
```

 SQS

SQS queue

arn:aws:sqs:us-east-2:██████████:POC-Queue

☒ Activate triggerProvisioned mode | [Info](#)☐ Configure☐ Enable EventCount metrics

Batch size - optional

10

☰ [Lambda](#) > [Functions](#) > POC-Lambda-1

CloudShell Feedback Console Mobile App

Lambda > Functions > POC-Lambda-1

CodeTestMonitorConfigurationAliasesVersions

Test eventInfo

To invoke your function without saving an event, configure the JSON event, then choose Test.

CloudWatch Logs Live TailSaveTest

Test event action

Create new eventEdit saved event

Invocation type

Synchronous
Executes the Lambda function and blocks until receiving the function's response, with a maximum timeout of 15 minutes. Returns function output or error details directly to the calling application.

Asynchronous
Enqueues the Lambda function for execution and returns immediately with a request ID. Function processes independently, with results optionally sent to a configured destination like SNS, SQS, or EventBridge.

Event name

POC-Lambda-Test-1

Maximum of 25 characters consisting of letters, numbers, dots, hyphens and underscores.

Event sharing settings

Private
This event is only available in the Lambda console and to the event creator. You can configure a total of 10. Learn more

Shareable
This event is shared by IAM users within the same account who have permissions to access and use shareable events. Learn more

Template - optional

SQS

Event JSON

Format JSON

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Event JSON

Format JSON

```
1 {  
2   "Records": [  
3     {  
4       "messageId": "19dd0b57-b21e-4ac1-bd88-01bbb068cb78",  
5       "receiptHandle": "MessageReceiptHandle",  
6       "body": "Hello from SQS!",  
7       "attributes": {  
8         "ApproximateReceiveCount": "1",  
9         "SentTimestamp": "1523232000000",  
10        "SenderId": "123456789012",  
11        "ApproximateFirstReceiveTimestamp": "1523232000001"  
12      },  
13      "messageAttributes": {},  
14      "md5OfBody": "{{{md5_of_body}}}",  
15      "eventSource": "aws:sqs",  
16      "eventSourceARN": "arn:aws:sqs:us-east-1:123456789012:MyQueue",  
17      "awsRegion": "us-east-1"  
18    }  
19  ]  
20 }
```

1:1JSON

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CodeTestMonitorConfigurationAliasesVersions

Executing function: succeeded(logs)

Details

{
 "statusCode": 200,
 "body": "\\Hello from Lambda!\\\\"
}

Summary

Code SHA-256
HAPq9EReJVEc5gLavtc/gyd5vZtd9eiUGF932t0jBxY=

Function version
\$LATEST

Duration
1.59 ms

Resources configured
128 MB

Init duration
85.52 ms

Execution time
21 seconds ago

Request ID
72701659-6af5-4da6-8561-4e65bb08

Billed duration
88 ms

Max memory used
31 MB

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Task 4.5: Verifying that the Lambda function adds the test message to a database

Attributes

Add new attribute ▾

Attribute name	Value	Type	Remove
orderID - Partition key	e5a7cd07-cd99-431a-b8e5-187de1407f2c	String	
order	Hello from SQS!	String	Remove

Cancel

Save

Save and close

Task 5: Enabling DynamoDB Streams

DynamoDB

Dashboard

Tables

Explore items

PartiQL editor

Backups

Exports to S3

Imports from S3

Integrations

Reserved capacity

Settings

DAX

Clusters

Subnet groups

Parameter groups

Events

Filter by tag

Any tag key

Filter by tag value

Any tag value

Find tables

orders

Indexes

Monitor

Global tables

Backups

Exports and streams

Permissions

Exports to S3

info

Last updated

May 24, 2026, 03:40 (UTC+1:00)

View details

Export to S3

Showing all export jobs from the last 90 days.

Find exports

0 matches

Initiated by = User

Clear filters

Export ARN

Destination S3 bucket

Status

Export job start time (UTC+01:00)

Export type

No exports

Export to S3

DynamoDB stream details

Turn off

Stream status

On

View type

New image

Latest stream ARN

arn:aws:dynamodb:us-east-2:123456789012:table/orders/stream/2026-05-24T02:40:52.617

Resource-based policy

Not active

CloudShell

Feedback

Console Mobile App

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DynamoDB > Tables > orders > Turn on DynamoDB stream

Share your feedback on Amazon DynamoDB

Your feedback is an important part of helping us provide a better customer experience. Take this short survey to let us know how we're doing.

Share feedback

Turn on DynamoDB stream

DynamoDB stream details

Capture item-level changes in your table, and push the changes to a DynamoDB stream. You then can access the change information through the DynamoDB Streams API.

View type

Choose which versions of the changed items you would like to push to the DynamoDB stream.

☐ New and old images

Both the new and old images of the changed item.

☐ Key attributes only

Only the key attributes of the changed item.

☒ New image

The entire item as it appears after it was changed.

☐ Old image

The entire item as it appears before it was changed.

Cancel

Turn on stream

Task 6.1: Creating a topic in the notification service

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New Feature
Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

Create subscription

Details

Topic ARN

arn:aws:sns:us-east-2:██████████:POC-Topic

Protocol

The type of endpoint to subscribe

Email

Endpoint

An email address that can receive notifications from Amazon SNS.

██████████@gmail.com

After your subscription is created, you must confirm it. [Info](#)

Subscription filter policy - optional [Info](#)
This policy filters the messages that a subscriber receives.

Redrive policy (dead-letter queue) - optional [Info](#)
Send undeliverable messages to a dead-letter queue.

Amazon SNS

- Dashboard
- Topics
- Subscriptions
- ▼ Mobile
 - Push notifications
 - Text messaging (SMS)

Subscription to POC-Topic created successfully.
The ARN of the subscription is arn:aws:sns:us-east-2:370047118847:POC-Topic:bdaba1fa-21b1-448b-b1e8-c44c0816da0e.

Notifications 0 0 0 1 0 1 0 0

Subscription: bdaba1fa-21b1-448b-b1e8-c44c0816da0e

Edit Delete

Details

ARN
arn:aws:sns:us-east-2:██████████:POC-Topic:bdaba1fa-21b1-448b-b1e8-c44c0816da0e

Endpoint
██████████@gmail.com

Topic
[POC-Topic](#)

Status
Pending confirmation

Protocol
EMAIL

[Subscription filter policy](#) Redrive policy (dead-letter queue)

Subscription filter policy [Info](#)

This policy filters the messages that a subscriber receives.

No filter policy configured for this subscription.
To apply a filter policy, edit this subscription.

Edit

Amazon SNS

- Dashboard
- Topics
- Subscriptions
- ▼ Mobile
 - Push notifications
 - Text messaging (SMS)

New Feature
Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

Subscription: bdaba1fa-21b1-448b-b1e8-c44c0816da0e

Edit Delete

Details

ARN
arn:aws:sns:us-east-2:██████████:POC-Topic:bdaba1fa-21b1-448b-b1e8-c44c0816da0e

Endpoint
██████████@gmail.com

Topic
[POC-Topic](#)

Status
Confirmed

Protocol
EMAIL

Lambda > Add triggers

Trigger configuration info

DynamoDB

aws database event-source-mapping nosql polling

DynamoDB table

Select or enter the ARN of a DynamoDB table. To select a stream in another AWS account, enter its DynamoDB stream Amazon Resource Name (ARN).

arn:aws:dynamodb:us-east-2:123456789012:table/orders

Event poller configuration

☒ Activate trigger

Select to activate the trigger now. Keep unchecked to create the trigger in a deactivated state for testing (recommended).

☐ Enable EventCount metrics

Track the number of events polled, filtered, invoked, and dropped by your event source mapping. CloudWatch charges apply.

Batch size

The maximum number of records in each batch to send to the function.

100

Starting position

The position in the stream to start reading from.

Latest

Batch window - optional

The maximum amount of time to gather records before invoking the function, in seconds.

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Task 7.3: Configuring the second Lambda function

Code source info

Open in Visual Studio Code

Update

POC-Lambda-2

lambda_function.py

```
1 import boto3, json
2
3 client = boto3.client('sns')
4
5 def lambda_handler(event, context):
6
7     for record in event["Records"]:
8
9         if record['eventName'] == 'INSERT':
10             new_record = record['dynamodb']['NewImage']
11             response = client.publish(
12                 TargetArn='<Enter Amazon SNS ARN for the POC-Topic>',
13                 Message=json.dumps({'default': json.dumps(new_record)}),
14                 MessageStructure='json'
15             )
```

EXPLORER

POC-LAMBDA-2

lambda_function.py

DEPLOY

Deploy (⌘%U)

Test (⌘%T)

TEST EVENTS [NONE SELECTED]

Create new test event

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Task 7.4: Testing the POC-Lambda-2 Lambda function

☰

Lambda > Functions > POC-Lambda-2

🔍

🔧

☑️ Successfully updated the function "POC-Lambda-2".

Code

Test

Monitor

Configuration

Aliases

Versions

Test event Info

CloudWatch Logs Live Tail

Save

Test

Test event action

Create new event

Edit saved event

Invocation type

Synchronous

Executes the Lambda function and blocks until receiving the function's response, with a maximum timeout of 15 minutes. Returns function output or error details directly to the calling application.

Asynchronous

Enqueues the Lambda function for execution and returns immediately with a request ID. Function processes independently, with results optionally sent to a configured destination like SQS, SNS, or EventBridge.

Event name

POC-Lambda-Test-2

Maximum of 25 characters consisting of letters, numbers, dots, hyphens and underscores.

Event sharing settings

Private

This event is only available in the Lambda console and to the event creator. You can configure a total of 10. [Learn more](#)

Shareable

This event is available to IAM users within the same account who have permissions to access and use shareable events. [Learn more](#)

Template - optional

DynamoDB Update

🔄

Event JSON

Format JSON

☑️ The test event "POC-Lambda-Test-2" was successfully saved.

Notifications 0 0 2 0 0 0

Code

Test

Monitor

Configuration

Aliases

Versions

☑️ Executing function: succeeded [\(logs\)](#)

▼ Details

null

Summary

Code SHA-256

DAo0Vv56wYg+P1eMzWM9QfAJ08lQytu9B8R98G/EPUg=

Function version

\$LATEST

Duration

249.46 ms

Resources configured

128 MB

Init duration

475.03 ms

Execution time

21 seconds ago

Request ID

690348a3-3d7d-4833-82ab-43a7937cd715

Billed duration

725 ms

Max memory used

79 MB

Log output

The area below shows the last 4 KB of the execution log. [Click here](#) to view the corresponding CloudWatch log group.

START RequestId: 690348a3-3d7d-4833-82ab-43a7937cd715 Version: \$LATEST
END RequestId: 690348a3-3d7d-4833-82ab-43a7937cd715
REPORT RequestId: 690348a3-3d7d-4833-82ab-43a7937cd715 Duration: 249.46 ms Billed Duration: 725 ms Memory Size: 128 MB Max Memory Used: 79 MB Init Duration: 475.03 ms

☑️ CloudShell

Feedback

📱 Console Mobile App

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Task 8: Creating an API with Amazon API Gateway

Networking & Content Delivery

API Gateway

Create and manage APIs at scale

Amazon API Gateway is a fully managed service that makes it easy for developers to create, publish, maintain, monitor, and secure APIs.

Get started

Create a new API to begin exploring API Gateway. You can also import an external definition file into API Gateway.

[Create an API](#)

How it works



API Gateway enables you to connect and access data, business logic, and functionality from backend services such as workloads running on Amazon Elastic Compute Cloud (Amazon EC2), code running on AWS Lambda, any web application, or real-time communication applications.

Pricing

With Amazon API Gateway, you only pay when your APIs are in use. There are no minimum fees or upfront commitments. For HTTP and REST APIs, you pay based on API calls received and amount of data transferred out. For WebSocket APIs, you pay based on number of messages and connection duration.

[Learn more about pricing](#)

Resources

- [Getting Started Guide](#)
- [Developer Guide](#)
- [API References](#)

API Gateway > APIs > Create API > Create REST API

Create REST API [Info](#)

API details

☒ **New API**
Create a new REST API.

☐ **Clone existing API**
Create a copy of an API in this AWS account.

☐ **Import API**
Import an API from an OpenAPI definition.

☐ **Example API**
Learn about API Gateway with an example API.

API name

Description - optional

API endpoint type
Regional APIs are deployed in the current AWS Region. Edge-optimized APIs route requests to the nearest CloudFront Point of Presence. Private APIs are only accessible from VPCs.

Regional

Security policy - new [Info](#)
Transport Layer Security (TLS) protects data in transit between a client and server. The security policy also determines the cipher suite options that clients can use with your API.

Choose a security policy

IP address type [Info](#)
Select the type of IP addresses that can invoke the default endpoint for your API.

☒ **IPv4**
Supports only edge-optimized and Regional API endpoint types.

☐ **Dualstack**

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Method details

Method type

POST

Integration type

☐ Lambda function

Integrate your API with a Lambda function.



☐ HTTP

Integrate with an existing HTTP endpoint.



☐ Mock

Generate a response based on API Gateway mappings and transformations.



☒ AWS service

Integrate with an AWS Service.



☐ VPC link

Integrate with a resource that isn't accessible over the public internet.



AWS Region

us-east-2

AWS service

Simple Queue Service (SQS)

AWS subdomain

HTTP method

POST

Action type

Task 9: Testing the architecture by using API Gateway

API Gateway

APIs

Custom domain names

Domain name access associations

VPC links

AgentCore targets [New](#)

▼ API: POC-API

[Resources](#)

Stages

Authorizers

Gateway responses

Models

Resource policy

Documentation

Dashboard

API settings

Usage plans

API keys

Client certificates

Settings

Create resource

/

POST

Client

Method request

Integration request

Method response

Integration response

AWS integration

Method request

Integration request

Integration response

Method response

Test

Test method

Make a test call to your method. When you make a test call, API Gateway skips authorization and directly invokes your method.

Query strings

param1=value1¶m2=value2

Headers

Enter a header name and value separated by a colon (:). Use a new line for each header.

header1:value1

header2:value2

Client certificate

No client certificates have been generated.

Request body

1 { "item": "latex gloves",

2 "customerID": "12345" }

DynamoDB

Dashboard

Tables

[Explore items](#)

PartiQL editor

Backups

Exports to S3

Imports from S3

Integrations

Reserved capacity

Settings

▼ DAX

Clusters

Subnet groups

Parameter groups

Events

Tables (1)

Filter by tag

Any tag key

Filter by tag value

Any tag value

Find tables

1

orders

orders

▼ Scan or query items

Scan

Query

Select a table or index

Table - orders

Select attribute projection

All attributes

Filters - optional

Run

Reset

Completed · Items returned: 2 · Items scanned: 2 · Efficiency: 100% · RCUs consumed: 2

Table: orders - Items returned (2)

Scan started on May 24, 2026, 04:22:26

orderID (String) | order

e5a7cd07-cd99-431a... | Hello from SQS!

62c6ecde-72aa-4e88-... | { "item": "latex gloves", "customerID": "12345" }

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Your feedback is an important part of helping us provide a better customer experience. Take this short survey to let us know how we're doing.

Share feedback

Edit item

You can add, remove, or edit the attributes of an item. You can nest attributes inside other attributes up to 32 levels deep. [Learn more](#)

Form | JSON view

Attributes

Add new attribute

Attribute name

Value

Type

Remove

orderID - Partition key

62c6ecde-72aa-4e88-851e-69ffebc3646f

String

order

{ "item": "latex gloves", "customerID": "12345" }

String

Remove

Cancel

Save

Save and close